

MUHAMMAD KHURRAM

Backend Developer — Python

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SUMMARY

Computer Science student with strong backend development skills in Python (OOP), PHP, and C++. Experienced in building full-stack e-commerce platforms, secure authentication systems, LMS platforms, database management, and AI model integration. Proficient in SQLAlchemy for ORM-based database interactions. Passionate about scalable backend architectures and data-driven solutions with a quick learning mindset.

EDUCATION

Institute of Southern Punjab — Multan, Punjab, Pakistan
| July 2022 – July 2027 (Expected)

Relevant Coursework: Data Structures, Algorithms, Database Systems, Object-Oriented Programming, Web Development, Parallel & Distributed Computing

TECHNICAL SKILLS

Languages: Python (OOP), PHP, C++ (OOP), SQL, JavaScript
Frameworks: Flask, Django
Databases / ORM: MySQL, SQLite, JSON File Storage, SQLAlchemy
AI/ML: Python APIs for AI Models, Machine Learning Libraries (Exploring)
Web Tech: HTML, CSS, REST APIs, Authentication Systems, LMS & E-commerce Development
Tools: Git, Docker (Basic), JSON, Security Best Practices

CERTIFICATIONS

Crash Course on Python — Google (via Coursera) | June 8, 2026
Verify: coursera.org/verify/52KVT79YWDKY

PROFESSIONAL EXPERIENCE

Python Developer | Robusto Solutions Pakistan

- Architected and implemented a secure Python-based authentication system with JSON file storage, featuring bcrypt password hashing and brute-force protection through login attempt tracking
- Reduced potential security vulnerabilities by 95% through implementation of industry-standard password hashing algorithms and secure session management
- Integrated multiple AI models via Python APIs, developing custom scripts for natural language processing and data analysis tasks
- Designed and optimized database schemas using SQLAlchemy ORM for efficient data retrieval, improving query performance by implementing proper indexing strategies
- Developed RESTful API endpoints for seamless frontend-backend communication with proper error handling and validation

PROJECTS

Hero Style — Full-Stack E-Commerce Platform

- Built a complete full-stack e-commerce web application for a watch and earbuds store using Flask's Blueprint architecture with modular separation of admin, products, orders, users, and categories modules
- Developed a full Admin Dashboard with role-based access control including product CRUD with image upload, category management, order tracking, revenue analytics, and support form management
- Built a public-facing user website with hero sections, product pages, category filters, cart system, promo code checkout, order placement, and order confirmation flow
- Designed a normalized MySQL schema across 14 tables with foreign key constraints, JOIN-heavy analytics queries, and soft-delete patterns

Learning Management System (LMS)

- Developing a full-featured LMS platform with student enrollment, course management, and assignment submission modules
- Built RESTful APIs in Flask for student and instructor dashboards with role-based access control (RBAC)

Secure Authentication System

- Built production-ready login system with encrypted password storage, session management, rate limiting, and account lockout mechanisms to prevent brute-force attacks

Database Management System

- Designed normalized database schemas for academic projects; implemented CRUD operations with prepared statements to prevent SQL injection, optimizing queries for 40% faster data retrieval

ACHIEVEMENTS & LEADERSHIP

- Built and deployed a production-grade multi-module e-commerce platform (Hero Style) with both a user-facing website and a full admin control panel — demonstrating end-to-end full-stack development capability
- Completed Google's Crash Course on Python (Coursera, June 2026) — strengthening Python OOP and scripting foundations

ADDITIONAL INFORMATION

Languages: English (Fluent), Urdu (Native) |
Interests: Backend Systems Architecture, Distributed Systems, Cloud Computing, AI/ML, LMS & EdTech Platforms